00-F-545F
December 27, 1985
SUPERSEDING
00-F-545E
May 18, 1977

FEDERAL SPECIFICATION

FOOD BLENDERS, ELECTRIC

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two sizes of electrically powered food blenders.

1.2 Classification. Food blenders shall be of the following sizes, as specified (see 6.2).

Size 1 - 1 quart container capacity
Size 2 - 4 quart container capacity

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-P-291	- Paperboard, Wrapping and Cushioning.
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standard:

FED-STD-123	- Marking for Shipment (Civil Agencies).

FSC 7320

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(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications and Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal standards and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA ; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-V-173	-	Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-461		Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute, Inc., (ANSI)

S-1.13 - Methods for the Measurement of Sound Pressure Levels

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate Sheet and Strip
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Underwriters Laboratories Inc.

Standard No. 73 - Motor-Operated Appliances

(Application for copies should be addressed to Underwriters Laboratories, 333 Pfingsten Road, Northbrook, IL 60062.)

National Sanitation Foundation (NSF)

Standard No. 8 - Commercial Powered Preparation Equipment
Listing of Food Service Equipment

Criteria C-2 - Special Equipment and/or Devices

(Application for copies should be addressed to National Sanitation Foundation, 3475 Plymouth Rd., P.O. Box 1468, Ann Arbor, MI 48106.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The food blenders shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturers standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturers or

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commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair service on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The food blenders shall conform to the applicable requirements of NSF Standard No. 8 or Criteria C-2 and to UL Standard No. 73. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.3).

3.3 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial food blenders provided the completed item complies with all provisions of this specification.

3.3.1 Stainless steel. Stainless steel shall conform to ASTM A 167.

3.4 Design and construction. The food blender shall consist essentially of a base, a container, a blending assembly, an electric drive motor, and a switch. The blending assembly shall be joined to the bottom of the container and be easily removable for cleaning or repair.

3.4.1 Base. The base shall be fabricated from metal or plastic and shall support the motor and container. The upper part of the base shall hold and support the container.

3.4.2 Container. The container for the size 1 blender shall have a capacity not less than 30 fluid ounces and the container for the size 2 blender shall have a capacity not less than 126 fluid ounces (see 4.4.3). The container for the size 1 blender shall be fabricated of stainless steel not thinner than 0.031 inch, or heat resistant glass. The container for the size 2 blender shall be fabricated of minimum 0.031 inch thick stainless steel and shall be provided with a loop handle formed from a single piece of stainless steel. Both size containers shall have a removable unthreaded splash cover of stainless steel or of plastic.

3.4.3 Blending assembly. The blending assembly shall consist of a stainless steel blade and drive shaft. The blade shall have a minimum of four separate sharpened cutting edges. The drive shaft assembly, mounted in the container shall be leakproof. The blade and drive shaft shall have a Rockwell hardness of 34 to 44 when measured on the "C" scale (see 4.4.1).

3.4.4 Electrical. The electric motor shall be the universal type. Unless otherwise specified (see 6.2), the motor shall operate on a nominal 120 volt, 60 hertz, single phase power source. The motor shall have permanently lubricated bearings. The blender shall be completely wired and equipped with a three-conductor cord, of 6-foot minimum length, and a three-prong grounding plug.

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3.4.4.1 Electromagnetic compatibility. When specified (see 6.2), the blender shall be designed and equipped for electromagnetic compatibility in accordance with class B of MIL-STD-461.

3.4.4.2 Switch. A control switch shall be located on the housing. Size 1 blender switch shall have a minimum of three positions: "off" and "high" with at least one intermediate speed. Size 2 blender switch shall have a minimum of four positions: "off" and "high" with at least two intermediate positions. All switch positions shall be clearly indicated.

3.4.4.3 Fungus and moisture resistance of electrical components. When specified (see 6.2), food blenders shall be treated for fungus and moisture resistance. The electrical circuitry including all components and connections shall be protected from the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173 with the following exceptions:

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coatings shall not be treated.

Varnish shall have a minimum dry-film thickness of 0.001 inch on component or element surfaces previously cleaned and prepared.

3.4.5 Noise limits. The food blenders shall not exceed 85 decibels A-weighted at the normal operator position when tested as specified in 4.4.5.

3.5 Operation. The food blender shall operate without undue vibration, heating, or noise when tested as specified in 4.4.2.

3.6 Repair and maintenance. Provisions shall be made to allow for ready adjustment, service, and replacement of electrical components and operational electrical accessories without disturbing or disassembling any other major component. There shall be free access to inspect, service, and adjust any other component or equipment, as required, for proper operation of the blender.

3.7 Finish. Finish of the food blender and container shall be in accordance with applicable requirements of NSF Standard No. 8.

3.8 Identification marking. Identification marking shall be permanently and legibly marked directly on the item or an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved, or applied by photo sensitive means. The plate thickness shall be not less than 0.012 inch. The identification plate shall include marking requirements of UL 73, and shall include blender capacity, manufacturer's name, contract number, and national stock number. An identification plate made of polyester laminate, 0.005 inches thick, with a pressure-sensitive backing is also acceptable.

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3.9 Workmanship. All components and assemblies of the food blender shall be free from dirt and other extraneous materials; burrs; slivers; rough die, tool and grind marks; dents; and cracks. Castings, moulded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes, and sprues.

3.9.1 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts, or incomplete fusion. All scale and flux shall be removed from the finished weld area.

3.9.2 Threaded fasteners. Threaded fasteners shall not be missing, broken, cracked, or stripped.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 In-process visual examination. When fungus and moisture resistance treatment is specified (see 6.2), an examination shall be made to establish that no deviations are made from the requirements specified in 3.4.4.3. When non-conformance is noted, correction shall be made to the affected items and process.

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4.2.2.2 In-process hardness testing. The blade and drive shaft shall be tested for hardness in accordance with 4.4.1. The lot size shall be expressed in units of one blade and drive shaft. The sample unit shall be one blade and drive shaft. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

4.2.3 End item visual examination. The end item shall be examined for defects listed in table I. The lot size shall be expressed in units of food blenders. The sample unit shall be one food blender. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements.	X	
Design, construction, and workmanship, general	Component fractured, split punctured, sprung, or malformed.	X	
	Any burr, sliver, or sharp edge.	X	
	Not mounted or joined as specified.	X	
	Components not accessible for repair and maintenance.	X	
Blending assembly	Not specified number of cutting edges on blade.	X	
	Blade not sharpened as specified.	X	
	Blending assembly not removable.	X	
Base	Does not hold and support the container.	X	
Container	Cover not type specified.	X	
	Handle not provided (when applicable).	X	
Welding and brazing (when applicable)	Missing, through burn holes, cracked, severe undercut, porous, fractured, or otherwise not fused; not ground smooth.	X	
	Slight slag inclusion or slight undercut.		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Finish	Not as specified.	X	
Threaded components	Missing, broken, stripped, or fractured.	X	
Electrical	Cord and plug do not provide grounding.	X	
	Switch positions not in specified order.	X	
	Switch not clearly marked.	X	
Marking identification, instructions	Missing, incomplete, not legible, or not specified type.	X	

4.2.4 End item dimensional examination. The end item shall be examined for conformance with the minimum cord length specified in 3.4.4. Any cord length not meeting the specified minimum shall be classified a defect. The lot size shall be expressed in units of food blenders. The sample unit shall be one food blender. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.2.5 End item testing.

4.2.5.1 Operational testing. Every end item shall be tested as specified in 4.4.2. Any end item exhibiting one or more test failures shall be rejected.

4.2.5.2 Representative item testing. One representative end item from the lot shall be tested as specified in 4.4.3 and 4.4.5 and as specified in 4.4.4 when applicable. Any test failure shall be cause for rejection of the lot.

4.2.6 Packaging inspection. An inspection shall be made to determine that the preservation, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size location, sequence, or method of application.
Materials	Any component missing, damaged, defective, or not as specified.

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Examine	Defect
Contents	Number per container not as specified.
Workmanship	Components not snug fit in container (loose).

4.2.7 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.3 Certification compliance examination. Certifications, certified test reports, or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements (see 3.2), shall be examined and validated as proof of compliance (see 4.3.1 and 4.3.2).

4.3.1 NSF. Acceptable evidence of meeting the requirements of NSF Standard No. 8 or Criteria C-2, shall be one of the following:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished blender.

b. A certification for the blender issued by NSF under their special one-time contract evaluation/certification service, or

c. A certified test report, acceptable to the contracting officer with the advice of the Army Surgeon General from an independent testing laboratory, indicating that the blender has been tested and conforms to NSF Standard No. 8 or Criteria C-2.

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4.3.2 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the food blenders and applicable components offered have been tested and conform to UL Standard No. 73.

4.4 Methods of inspection.

4.4.1 Rockwell hardness test. The blade and the drive shaft shall be tested for hardness in accordance with ASTM E 18. Nonconformance to the hardness requirements of 3.4.3 shall constitute failure of this test.

4.4.2 Operational test. Food blenders shall be operated without load at all speeds to determine compliance with the requirements specified in 3.5. Excessive vibration, heat, or noise shall constitute failure of this test.

4.4.3 Capacity test. The containers shall be filled to capacity with fluid and the volume of fluid measured. Failure to hold the volume of fluid specified in 3.4.2 shall constitute failure of this test.

4.4.4 Electromagnetic compatibility test. When electromagnetic compatibility is required, the food blender shall be tested in accordance with Methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

4.4.5 Noise level test. The food blender noise level shall be measured in accordance with the filled method of ANSI S-1.13 except that the A-weighted network shall be used in lieu of the octave or narrow band analyzers to determine conformance with 3.4.5. Any nonconformance to be within the noise limit shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each food blender shall be unit packed in a snug-fitting fiberboard box conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Blenders shall be completely wrapped with cushioning material conforming to type II, class A of PPP-C-843, or single-faced corrugated fiberboard conforming to type optional, style 1 of PPP-P-291, and the wrap secured with tape conforming to type III, grade A of PPP-T-45; or cushioned and immobilized within the box by means of buffer pads, cell spacers, or die-cut pads formed from scored sheets of the same material as the box. Each box shall be securely closed with 2-inch minimum width paper tape as specified for the wrap.

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5.1.2 Commercial. Each type food blender shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Six 1-quart or two 4-quart food blenders, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Six 1-quart or two 4-quart food blenders, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in PPP-B-636.

5.2.3 Commercial packing. Food blenders, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), food blenders packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

5.4.2 Military requirements. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

6.1 Intended use. The electric food blenders are intended for use in blending and mixing operations in kitchens and laboratories.

6.2 Ordering data. Purchasers should exercise the preferred options permitted herein and include the following information in procurement documents.

a. Title, number, and date of this specification.
b. Size of blender required (see 1.2).
c. Electrical characteristics, if other than specified (see 3.4.4).
d. When electromagnetic compatibility is required (see 3.4.4.1).
e. When fungus and moisture resistance treatment is required (see 3.4.4.3).
f. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
g. When weather-resistant fiberboard containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).
i. Marking required (see 5.4).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3).

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD, TS
Navy - MS
Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES

GSA - FSS
VA - DMS

PREPARING ACTIVITY:

Army - GL

Project No. 7320-0765

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